THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS UNIT TEST SET A

Chapters 1–4: **Orienting Yourself: The Use of Coordinates, Introduction to Linear Polynomials, The World of Numbers, Exploring Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The distance between $A(-1,\ -1)$ and $B(4,\ 11)$ is: 1

(a) 12 units **(b)** 14 units **(c)** 13 units **(d)** 15 units

2. The point $(4,\ -7)$ lies: 1

(a) Quadrant III **(b)** Quadrant II **(c)** Quadrant IV **(d)** Quadrant I

3. The midpoint of the segment joining $A(5,\ 9)$ and $B(-1,\ -3)$ is: 1

(a) $(-3,\ -6)$ **(b)** $(2,\ -3)$ **(c)** $(4,\ 6)$ **(d)** $(2,\ 3)$

4. The degree of the polynomial $y^3 + 2y - 1$ is: 1

(a) 3 **(b)** 2 **(c)** 4 **(d)** 5

5. The sequence $10, 7, 4, 1, \ldots$ forms a linear pattern. Its nth term is: 1

(a) $10n - 3$ **(b)** $-3n + 13$ **(c)** $-3n + 10$ **(d)** $-3n - 7$

6. The graph of $y = -2x$ rises or falls as x increases? 1

(a) falls (negative slope) **(b)** is vertical
(c) is horizontal **(d)** rises

7. The decimal expansion of $\frac{5}{6}$ is: 1

(a) non-terminating non-repeating **(b)** non-terminating repeating
(c) terminating **(d)** cannot be determined

8. Who gave the first proof of the irrationality of $\sqrt{2}$ by contradiction (c. 400 BCE)? 1

(a) Lambert **(b)** Hippasus **(c)** Āryabhaṭa **(d)** Brahmagupta

9. The number $\sqrt{81}$ is: 1

(a) not a real number **(b)** irrational
(c) neither rational nor irrational **(d)** rational

10. Expand $(3x + 4)^2$: 1

(a) $9x^2 + 16$ **(b)** $9x^2 + 12x + 16$ **(c)** $9x^2 + 24x - 16$ **(d)** $9x^2 + 24x + 16$

11. $(7x + 2)(7x - 2)$ equals: 1

(a) $49x^2 + 28x + 4$ **(b)** $49x^2$ **(c)** $49x^2 - 4$ **(d)** $49x^2 - 28x + 4$

12. The expression $4s^2 + 20st + 25t^2$ can be factorised as: 1

(a) $(4s + 5t)^2$ **(b)** $(2s + 25t)^2$ **(c)** $(2s - 5t)^2$ **(d)** $(2s + 5t)^2$

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

13. Check whether $P(1,\ 0)$ is equidistant from $A(2,\ 3)$ and $B(4,\ -1)$. 2

14. Find the values of the linear polynomial $4x - 1$ at $x = 3$ and $x = 0$. 2

15. A linear pattern begins $4,\ 7,\ 10, \ldots$ Write the expression for its nth term and find the 10th term. 2

16. Without long division, state whether $\frac{5}{9}$ has a terminating or a repeating decimal. Give a reason. 2

17. Expand $\left(\frac{5p}{2} + \frac{3q}{2}\right)^2$ using a suitable identity. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. Show that $A(-2,\ -1)$, $B(3,\ -1)$, $C(3,\ 2)$, $D(-2,\ 2)$ are the vertices of a rectangle. 3

19. The cost of a journey is $C(d) = 100 + 60d$. Find the cost of 15 km and the distance for a cost of ₹700. Is it linear growth or decay? 3

20. Express $2.45\overline{37}$ in the form $\frac{p}{q}$. 3

21. Factorise $8n^3 - 60n^2m + 150nm^2 - 125m^3$. 3

22. Factorise $9x^2 + 12xy + 6xz + 4y^2 + 4yz + z^2$. 3

OR

Simplify $\dfrac{p^4 - 16}{p^2 - 4p + 4}$ $(p \neq 2)$. 3

23. Show that the points $M(-3,\ -4)$, $A(0,\ 0)$ and $G(6,\ 8)$ are collinear. 3

OR

Show that the triangle with vertices $A(0,\ 0)$, $B(6,\ 0)$, $C(3,\ 4)$ is isosceles. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. A mobile phone costs ₹10,000 and loses ₹800 in value every year. 5

(i) Find its value after 3 years.
(ii) Make a table for $t = 0$ to 8 years.
(iii) Write v in terms of t and explain why this is linear decay.
(iv) After how many years will its value be ₹2,000?

OR

Case study: A growing pattern of tiles has 1, 3, 5, 7, … tiles in stages 1, 2, 3, 4. 5

(i) Find the expression for the number of tiles in stage n.
(ii) How many tiles are there in stages 15 and 26?
(iii) Which stage has 21 tiles? Which has 47?
(iv) What is the slope of the line $y = 2x - 1$ and what does it represent?

25. **Case study:** Saira arranges 1 square of side x, 8 rectangular strips of $x \times 1$ and 15 unit squares into a rectangle. A rectangular pool has breadth 4 m less than its length and area 96 m². 5

(i) Find Saira's rectangle's length and breadth.
(ii) Form and solve the equation for the pool.
(iii) Find the pool's perimeter.

OR

(i) Factorise $9a^2 + b^2 + 4c^2 - 6ab + 12ac - 4bc$. 5

(ii) Factorise $16s^2 + 25t^2 - 40st$.

END OF QUESTION PAPER — **Total: 50 Marks**